

Normalisation Worked Example — 1NF to 3NF

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Overview

Normalisation is the process of restructuring a flat, denormalised table into a set of focused, well-organised tables that eliminate redundancy and prevent update anomalies. As Chapter 12 explains, these anomalies are exactly the kind of silent data corruption that produces wrong analytical results — the worst outcome for any analyst.

This walkthrough takes a single messy flat table through First Normal Form (1NF), Second Normal Form (2NF), and Third Normal Form (3NF), showing the exact decisions made at each step.

The Starting Point: A Denormalised Flat Table

Imagine a retail company stores all order information in one table called `orders_flat`.

order_id	customer_id	customer_name	customer_city	product_ids	product_names	unit_prices	quantity	order_date
1001	42	Sarah Chen	Austin	P01, P03	Notebook , Stapler	3.99, 1.49	2, 1	2024-01-15
1002	17	Marcus Webb	Denver	P02	Pen Set	5.99	3	2024-01-16
1003	42	Sarah Chen	Austin	P01	Notebook	3.99	5	2024-01-18

Problems visible immediately:

- `product_ids`, `product_names`, `unit_prices`, and `quantity` hold multiple values in a single cell (comma-separated lists).
- `customer_name` and `customer_city` repeat every time customer 42 places an order.
- Changing Sarah Chen's city requires updating every row she appears in — a classic update anomaly.

Step 1 — Apply First Normal Form (1NF)

Rule: Every column must hold a single, atomic value. No repeating groups.

Split each product onto its own row so every cell contains exactly one value.

order_id	customer_id	customer_name	customer_city	product_id	product_name	unit_price	quantity	order_date
1001	42	Sarah Chen	Austin	P01	Notebook	3.99	2	2024-01-15
1001	42	Sarah Chen	Austin	P03	Stapler	1.49	1	2024-01-15
1002	17	Marcus Webb	Denver	P02	Pen Set	5.99	3	2024-01-16
1003	42	Sarah Chen	Austin	P01	Notebook	3.99	5	2024-01-18

The composite primary key is now (order_id, product_id) — the combination that uniquely identifies each row.

Redundancy still exists, but the table is now in 1NF.

Step 2 — Apply Second Normal Form (2NF)

Rule: Every non-key column must depend on the whole primary key, not just part of it.

With the composite key (order_id, product_id), ask: does each column need both parts to be determined?

- customer_id, customer_name, customer_city, order_date depend only on order_id — partial dependency.
- product_name, unit_price depend only on product_id — partial dependency.
- quantity depends on both order_id and product_id — correctly placed.

Solution: Split into three tables.

orders

order_id	customer_id	order_date
1001	42	2024-01-15
1002	17	2024-01-16
1003	42	2024-01-18

order_items

order_id	product_id	quantity
1001	P01	2
1001	P03	1
1002	P02	3
1003	P01	5

products

product_id	product_name	unit_price
P01	Notebook	3.99
P02	Pen Set	5.99
P03	Stapler	1.49

Customer data still lives inside orders. That leads to Step 3.

Step 3 — Apply Third Normal Form (3NF)

Rule: No non-key column should depend on another non-key column (no transitive dependencies).

In orders, customer_name and customer_city depend on customer_id, not directly on order_id. That is a transitive dependency through a non-key column.

Solution: Extract a customers table.

customers

customer_id	customer_name	customer_city
42	Sarah Chen	Austin
17	Marcus Webb	Denver

Update orders to reference customer_id as a foreign key only — the name and city now live in exactly one place.

The Fully Normalised Schema

Four focused tables — customers, orders, order_items, products — connected by primary and foreign keys. This is the schema blueprint Chapter 12 describes: every meaningful JOIN you write traces back to these key relationships. Updating Sarah Chen's city now requires changing exactly one row in customers. No anomalies. No corrupted analysis.

Analyst tip: When you encounter a slow or confusing query in production, check whether the schema reached 3NF. Denormalised tables often signal missing indexes too — another Chapter 12 warning sign that a two-second query can become a two-minute bottleneck.